

BALMACEDA, ChileWMO: **858740**Lat: **45.9128S**Lon: **71.6942W**Elev: **1693**StdP: **13.82**Time Zone: **-4.00 (CHL)**Period: **96-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	17.4	8.5	9.1	12.5	14.2	12.0	20.9	35.0	41.7	32.0	40.7	1.3	260	0.927

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.2	78.6	58.2	73.5	56.3	69.5	54.6	59.3	75.1	57.4	70.4	55.6	66.9	15.0	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
52.7	62.9	62.6	51.1	59.2	61.5	49.6	56.1	60.3	26.7	76.4	25.3	70.7	24.2	67.0	66.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.8	32.9	30.2	4.4	82.8	7.1	5.4	-0.7	86.7	-4.9	89.8	-8.8	92.8	-14.0	96.7		
(5)			3.9	62.0	7.1	2.1	-1.3	63.5	-5.4	64.8	-9.4	66.0	-14.6	67.5	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.6	54.6	54.2	50.4	45.1	39.4	34.6	33.8	37.1	41.2	45.1	48.1	52.4	(6)	(7)
	DBStd	9.10	5.66	5.52	5.11	5.50	6.28	6.91	7.37	5.24	4.63	4.42	4.87	5.67	(7)	(8)
	HDD50	2515	20	17	56	165	333	462	502	401	266	161	91	41	(8)	(9)
	HDD65	7454	325	306	452	598	795	912	967	866	715	617	508	393	(9)	(10)
	CDD50	541	162	133	69	17	3	0	0	0	1	9	33	114	(10)	(11)
	CDD65	5	2	2	0	0	0	0	0	0	0	0	0	1	(11)	(12)
	CDH74	388	145	141	19	0	0	0	0	0	0	0	10	73	(12)	(13)
	CDH80	98	43	42	1	0	0	0	0	0	0	0	1	11	(13)	(14)
	WSAvg	15.3	19.4	17.2	15.2	14.7	11.5	11.9	11.0	12.5	13.9	16.8	19.3	19.9	(14)	(15)
	PrecAvg	22.8	1.0	0.9	1.5	2.0	3.4	3.1	3.2	2.4	1.6	1.3	1.2	1.1	(15)	(16)
(16) Precipitation	PrecMax	37.4	4.1	3.0	4.6	7.2	13.5	8.8	8.8	5.9	5.9	5.8	4.8	3.1	(16)	(17)
	PrecMin	10.9	0.0	0.0	0.1	0.1	0.4	0.0	0.4	0.4	0.1	0.0	0.0	0.0	(17)	(18)
	PrecStd	5.9	0.9	0.7	1.0	1.6	2.3	2.0	2.0	1.2	1.1	1.0	0.9	0.8	(18)	(19)
															(19)	(20)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.9	86.1	76.7	68.0	58.8	50.4	50.5	53.2	63.0	67.7	75.0	81.3	(20)	(21)
		MCWB	61.7	59.6	57.5	53.6	50.5	44.5	45.0	44.8	46.9	50.8	54.8	60.3	(21)	(22)
	2%	DB	77.2	78.6	70.4	62.6	54.4	48.0	47.6	49.7	56.6	62.4	68.0	75.0	(22)	(23)
		MCWB	58.0	58.0	54.2	51.7	48.0	43.3	43.6	43.6	45.1	48.5	51.7	57.2	(23)	(24)
	5%	DB	71.8	71.9	66.0	58.9	51.7	46.0	44.8	47.2	53.3	58.4	62.5	69.2	(24)	(25)
		MCWB	56.1	56.3	52.9	50.2	46.5	42.0	41.2	41.9	44.2	47.4	50.4	54.6	(25)	(26)
	10%	DB	67.3	66.4	62.2	55.6	48.6	43.7	42.8	44.9	50.1	55.2	58.7	64.4	(26)	(27)
		MCWB	54.7	54.2	51.5	48.4	44.4	40.3	39.4	40.3	42.6	45.9	48.7	52.3	(27)	(28)
	0.4%	WB	62.0	61.8	59.6	55.7	51.6	45.8	46.1	47.1	49.1	52.4	55.8	60.8	(28)	(29)
		MCDB	82.7	79.9	72.0	64.2	56.6	49.2	49.7	50.7	59.9	64.3	70.9	78.8	(29)	(30)
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	59.0	59.2	56.0	53.4	49.1	43.7	43.7	44.5	46.5	49.6	53.2	57.8	(30)	(31)
		MCDB	74.3	75.0	66.7	60.3	53.4	46.9	46.8	48.6	54.7	59.6	65.5	72.3	(31)	(32)
	5%	WB	57.0	57.1	53.9	50.9	47.0	41.9	41.6	42.3	44.8	48.1	51.1	55.2	(32)	(33)
		MCDB	69.7	69.6	63.6	57.5	51.1	44.9	44.4	46.1	51.5	57.0	61.2	67.6	(33)	(34)
	10%	WB	54.9	55.0	52.2	48.9	44.9	40.2	39.4	40.6	43.1	46.4	49.1	53.0	(34)	(35)
		MCDB	65.5	65.2	61.1	54.9	48.5	43.2	42.0	44.2	49.3	54.3	57.5	63.1	(35)	(36)
	5% DB	MDBR	20.2	21.2	21.1	18.3	14.1	12.1	12.1	13.6	16.9	18.3	18.6	19.2	(36)	(37)
		MCDBR	33.3	35.1	34.5	26.2	17.3	13.1	12.8	16.4	23.3	26.6	28.3	31.2	(37)	(38)
		MCWBR	17.3	18.0	20.1	17.2	12.4	9.9	9.3	11.2	13.9	15.6	15.8	16.7	(38)	(39)
		MCWBR	27.4	28.7	27.0	22.9	15.1	11.5	11.6	14.4	20.3	21.8	23.9	27.1	(39)	(40)
(40) Clear-Sky Solar Irradiance	taub		0.306	0.308	0.297	0.286	0.273	0.263	0.261	0.278	0.294	0.301	0.298	0.308	(40)	(41)
		taud	2.571	2.569	2.593	2.616	2.603	2.564	2.548	2.575	2.547	2.539	2.566	2.545	(41)	(42)
	Ebn at Noon		322	312	301	280	259	247	259	278	296	311	323	324	(42)	(43)
		Edh at Noon	32	30	27	22	18	16	18	22	27	31	32	34	(43)	(44)
(44) All-Sky Solar Radiation	RadAvg		2255	1902	1370	875	493	353	420	663	1081	1621	2060	2272	(44)	(45)
	RadStd		120	153	85	49	47	37	39	59	98	114	94	180	(45)	(46)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page